Explainable Clinical Decision Support System

Clinical Assessment Report

Patient Details

Patient Id	pat0010
Patient Name	pat10
Doctor Name	mitchel
Hospital Name	cmc
Disease	Heart Disease
Age	35
Gender	Male
Cp	Typical Angina
Trestbps	100
Chol	170
Fbs	No
Restecg	Normal
Thalach	250
Exang	No
Oldpeak	0.2
Slope	Downsloping
Ca	0
Thal	Reversible Defect

Prediction

Disease	heart
Result	No Heart Disease
Probability (%)	1.16
Recommendation	No significant indication of heart disease was detected. Maintain a healthy lifestyle, exercise regularly, and undergo routine medical check-ups.

SHAP Explanation

Major Vessels	-0.9427
Chest Pain Type	-0.8407

Thalassemia	-0.7002
ST Depression	-0.6094
Age	-0.4774

LIME Explanation

cp <= -0.17	-0.1279
-0.90 < oldpeak <= -0.47	-0.0988
trestbps <= -0.78	-0.0522
thal <= -0.89	-0.0288
age <= -0.71	-0.0246

Cross-Explainer Concordance Index (CECI)

CECI score (%)	84.5
Feature overlap (%)	80
Direction agreement (%)	100
Ranking agreement (%)	75
Consistency level	High

Attribution-to-Recourse Alignment (ARA)

SHAP alignment (%)	100.0
LIME alignment (%)	50.0
Mean ARA (%)	75.0
Recourse features	major vessels, st depression
Interpretation	ARA measures overlap between model explanations and hypothetical recourse features; it is not clinical validation.

Counterfactual and Sensitivity Scenarios

Scenario 1	Risk-increasing sensitivity; probability 14.67%; ca: [0, 2]
Scenario 2	Risk-increasing sensitivity; probability 14.67%; ca: [0, 3]
Scenario 3	Risk-increasing sensitivity; probability 12.28%; ca: [0, 1]
Scenario 4	Risk-increasing sensitivity; probability 56.66%; oldpeak: [0.2, 2.5], ca: [0, 2]
Scenario 5	Risk-increasing sensitivity; probability 53.46%; oldpeak: [0.2, 2.5], ca: [0, 1]

This report is generated for academic demonstration only and is not a medical diagnosis.